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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 05/03/2006 |
| **PI Name:** | Clark, Brian Ph.D. | | |
| **Project Title:** | Neuromuscular function and countermeasures to prolonged unweighting | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | bcckar01@syr.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 315-443-1411 |
| **Organization Name:** | Syracuse University | | |
| **PI Address 1:** | Exercise Science | | |
| **PI Address 2:** | 820 Comstock Avenue, RM 201 | | |
| **PI Web Page:** | | | |
| **City:** | Syracuse | **State:** | NY |
| **Zip Code:** | 13206 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | GSRP |
| **Start Date:** | 07/01/2004 | **End Date:** | 06/30/2006 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA HQ |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The purpose of this study is to evaluate the effect of unilateral lower limb suspension (ULLS) on neuromuscular function, and determine the efficacy of applied ischemia and neural training as countermeasures. Subjects undergo 4-weeks of ULLS during which one group serves as a control, and two groups receive countermeasures. Neuromuscular function (i.e. strength, fatigability etc.) is assessed before and after ULLS. This study will further elucidate disuse-induced alterations in neuromuscular properties, and determine the efficacy of countermeasures designed to diminish dysfunction following unloading. These findings are important to NASA's mission, as long-term spaceflight causes large reductions in neuromuscular function. | | |
| **Rationale for HRP Directed Research:** | | | |
| **Research Impact/Earth Benefits:** | | | |

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| **Task Progress:** | No progress report this period. |
| **Bibliography Type:** | Description: (Last Updated: ) |